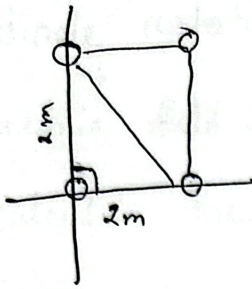
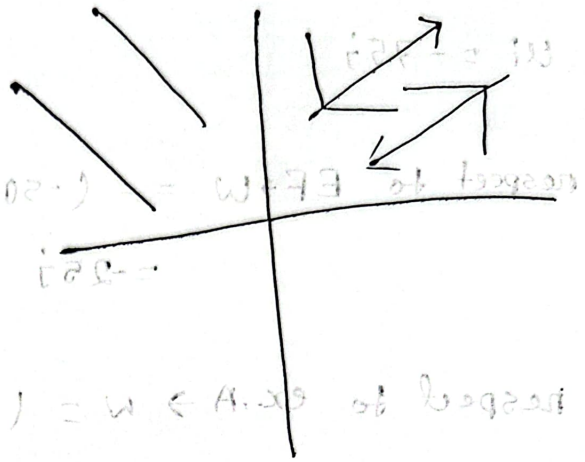


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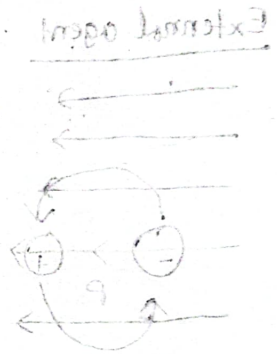
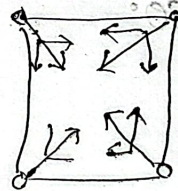
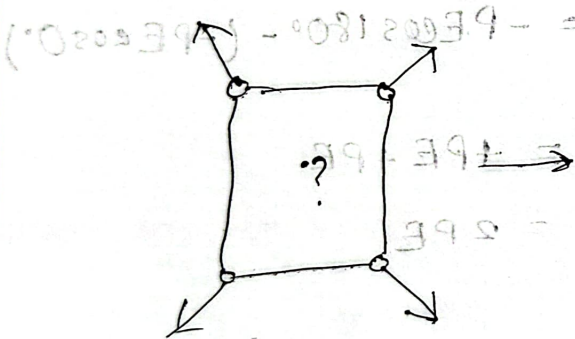


$$\cos \theta = \frac{2}{2\sqrt{2}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

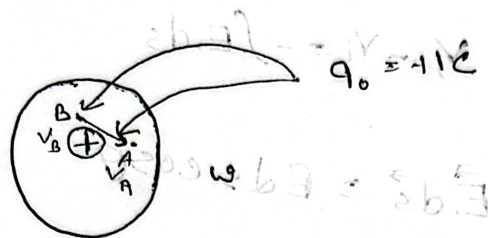


$$V - V = W$$



Electric Potential

potential due to a charge distribution



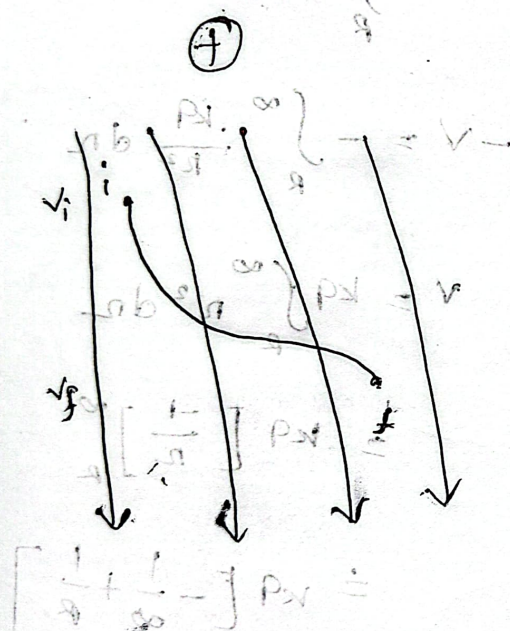
higher potential $\rightarrow$ lower
potential $\leftarrow$ higher

$$W = \int_A^B \vec{F} \cdot d\vec{s} = q_0 \int_A^B \vec{E} \cdot d\vec{s}$$

$$V_B - V_A = \frac{W}{q_0} = \frac{q_0 \int_A^B \vec{E} \cdot d\vec{s}}{q_0} = \int_A^B \vec{E} \cdot d\vec{s}$$

$$V_B - V_A = 0 \Rightarrow V_B = V_A$$

$$V = - \frac{W}{q_0} = - \frac{U}{q_0}$$



$$dW = \vec{F} \cdot d\vec{s}$$

$$= q \vec{E} \cdot d\vec{s}$$

$$\boxed{\vec{F} = q \vec{E}}$$

$$W = q \int_i^f \vec{E} \cdot d\vec{s}$$

$$-W = -q \int_i^f \vec{E} \cdot d\vec{s}$$

$$\Rightarrow -\frac{W}{q_0} = - \int_i^f \vec{E} \cdot d\vec{s}$$

$$\Delta V = - \int_i^f \vec{E} \cdot d\vec{s}$$

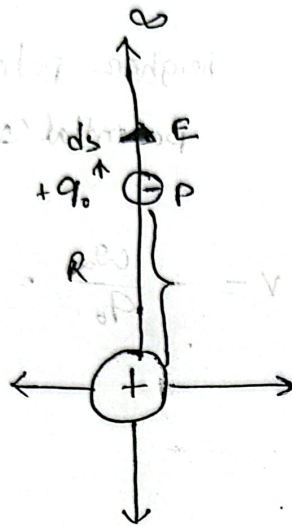
$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}$$

$$\int x^2 dx$$

3

Potential due to a charge Particle

Electrostatic potential



$$V_f - V_i = - \int E \cdot ds$$

$$\vec{E} \cdot d\vec{s} = E ds \cos \theta$$

$$E \cdot ds = E ds \cos \theta \text{ as } \theta = 0$$

$$= E dr$$

$$V_\infty - V = - \int_R^\infty E dr$$

$$\int x^n dx$$

$$\frac{1}{n+1}$$

$$\frac{1}{n+1}$$

$$\frac{1}{n+1}$$

$$\frac{1}{n+1}$$

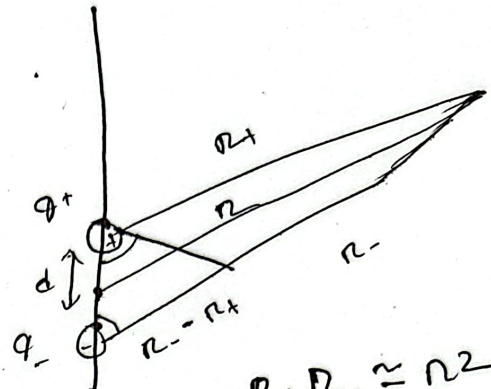
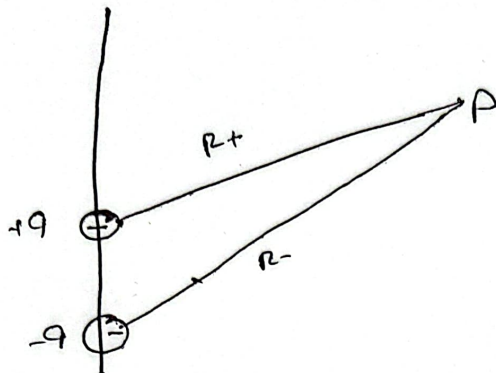
$$-V = - \int_R^\infty \frac{kq}{r^2} dr$$

$$V = kq \int_R^\infty \frac{1}{r^2} dr$$

$$= kq \left[-\frac{1}{r} \right]_R^\infty$$

$$= kq \left[-\frac{1}{\infty} + \frac{1}{R} \right]$$

P. D. to Electric dipole



$$V = \sum_{i=1}^2 V_i = V_+ + V_-$$

$$= \frac{1}{4\epsilon_0\pi} \cdot \frac{q}{r_+} + \frac{1}{4\epsilon_0\pi} \cdot \frac{-q}{r_-}$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_+} - \frac{1}{r_-} \right]$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{r_- - r_+}{r_+ r_-} \right]$$

$$r_+ r_- \approx r^2$$

$$\cos \theta \approx \frac{r_- - r_+}{d}$$

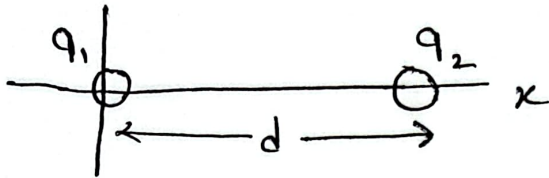
$$d \cos \theta \approx r_- - r_+$$

[negative value
 माना (-) value
 में आने 24]

$$V = \frac{q}{4\pi\epsilon_0} \cdot \frac{d \cos \theta}{r^2}$$

$$= \frac{q d \cos \theta}{4\pi\epsilon_0 r^2}$$

$$\therefore V = \frac{P \cos \theta}{4\pi\epsilon_0 r^2} \quad [P = qd]$$



$$\left\{ \begin{array}{l} q_1 = +5e \\ q_2 = -15e \\ d = 24.0 \text{ cm} \\ v = 0 \text{ at infinity} \end{array} \right.$$

What are the finite

(a) positive

(b) negative values of x at which the net

electric potential on the x -axis is zero?

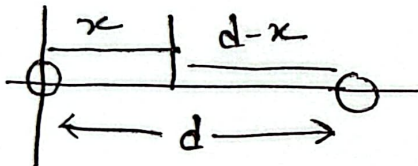
3 condition

~~$x < 0$~~

(i) $x < 0$ (ii) $x > d$

(iii) $0 < x < d$

(i)



$$v = v_1 + v_2 = 0$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{q_1}{x} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{d-x} = 0$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{x} + \frac{q_2}{d-x} \right) = 0$$

$$\Rightarrow \frac{q_1}{x} = \frac{-q_2}{d-x}$$

$$\Rightarrow \frac{5e}{x} = \frac{-(-15e)}{d-x}$$

$$\Rightarrow 5(d-x) = 15x$$

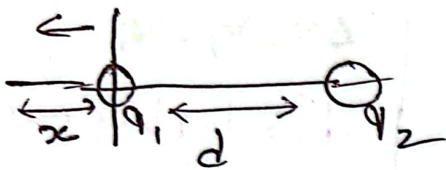
$$\Rightarrow d-x = 3x$$

$$\Rightarrow 3x = d-x$$

$$\Rightarrow 4x = d$$

$$\therefore x = \frac{d}{4} = \frac{24}{4} = 6 \text{ cm}$$

Point
[9, 12, 6 cm (5 cm) 12 value
3 cm (12 Point) Potential
value = 0]



$$V = V_1 + V_2 = 0$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{-x} + \frac{q_2}{-x+d} \right] = 0$$

$$\Rightarrow \frac{q_1}{-x} = \frac{-q_2}{x+d}$$

$$\Rightarrow \frac{5e}{-x} = \frac{-(-15e)}{-x+d}$$

$$\Rightarrow -15x = -5x + 5d$$

$$\Rightarrow -10x = 5d$$

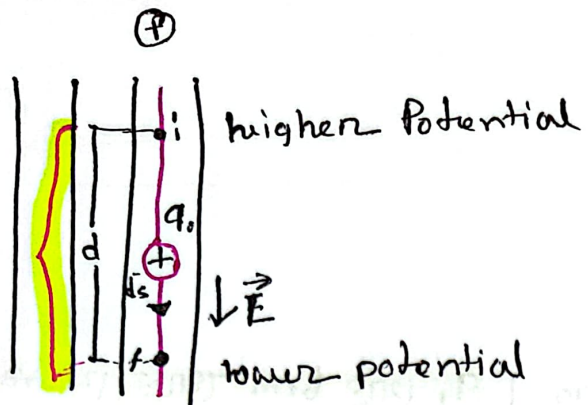
$$\Rightarrow -x+d = -3x$$

$$\Rightarrow -2x = d$$

$$\Rightarrow x = \frac{d}{-2} = \frac{24}{-2} = -12 \text{ cm}$$

[x is neg a
12 cm (12 Point)
[V = 0]]

#



Q_n i to f for reference
Potential ΔV change

$\vec{E} \cdot d\vec{s}$

$$\Delta V = - \int_i^f \vec{E} \cdot d\vec{s}$$

$$\Rightarrow - \int_i^f E ds$$

$$\Rightarrow - E \int_i^f ds$$

$$\therefore V_f - V_i = -Ed$$

Electric field
downward $\downarrow$

$\Delta V = V_f - V_i$

$d\vec{s} \rightarrow$ displacement
vector
($\vec{ds}$)

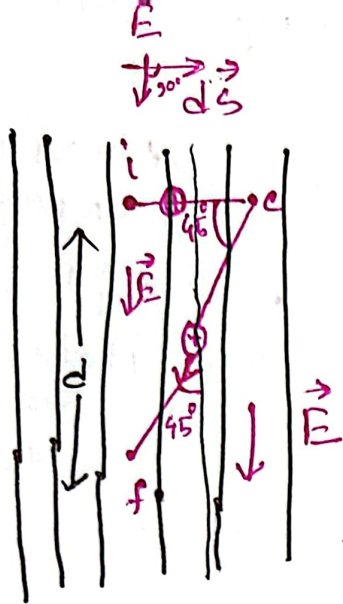
$\vec{A} \cdot \vec{B} = AB \cos \theta$

$\vec{E} \cdot d\vec{s} = E ds \cos 0^\circ$
 $= E ds$

ds is intg $\Rightarrow$ path length

#

Potential depends on displacement, But it
doesn't depend on the path



Qn : i to c to f a closed Potential do change

$$V_c - V_i = - \int_i^c E ds \cos 90^\circ$$

$$V_c - V_i = 0 \quad [i \text{ to } c \text{ shift is } \perp \text{ to } E \text{ Potential diff zero}]$$

$\frac{d}{\sqrt{2}}$

now,

$$V_f - V_c = - \int_c^f E ds \cos \theta$$

$$= - E \cos 45^\circ \int_c^f ds$$

$$= - E \cdot \frac{1}{\sqrt{2}} \cdot d\sqrt{2}$$

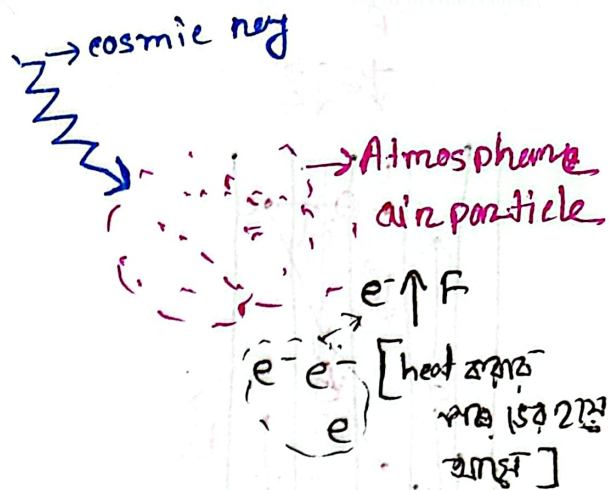
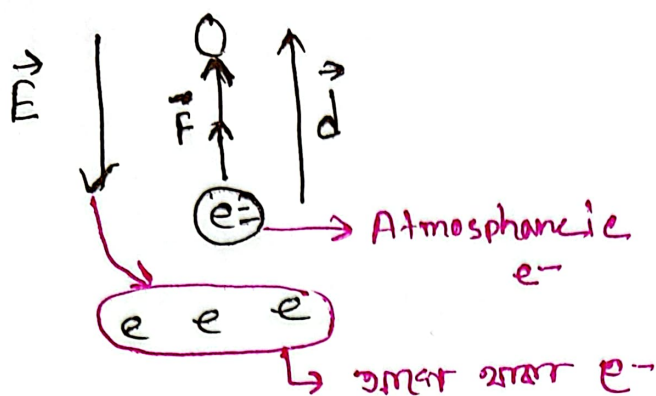
$$= - Ed$$

$$\begin{aligned} \sin 45^\circ &= \frac{if}{cf} \\ cf &= \frac{if}{\sin 45^\circ} \\ &= \frac{d}{\frac{1}{\sqrt{2}}} \\ &= d\sqrt{2} \end{aligned}$$

দুই point এর Potential difference (displacement) এর উপর depend করে। But কোন Path নিয়ে গুলো আমরা তার উপর depend করে না।

(Proved)

#



Q. ① What is the change ΔV in the electric potential energy of a released electron when the electric force cause it to move vertically upward through a distance $d = 520\text{m}$

$$E = 150\text{N/C}$$

$$d = 520\text{m}$$

② Through what potential change does the electric move?

③ From work done $\rightarrow$

$$W = \vec{F} \cdot \vec{d}$$

$$= q \vec{E} \cdot \vec{d}$$

from
Coulombs law / electrostatic F relation
 $q = \text{test charge } [-1.6 \times 10^{-19}]$

$$= e \vec{E} \cdot \vec{d}$$

$$[E \downarrow d \uparrow \theta] 180^\circ$$

$$= qEd \cos 180^\circ$$

$$= (-1.6 \times 10^{-19}) (150) (520) (-1)$$

$$= 1.2 \times 10^{-14} \text{ J [work done]}$$

now,

$$\Delta U = -W \cdot \left[\begin{array}{l} \text{relation between work done and} \\ \text{potential energy} \end{array} \right]$$

$$\Delta U = -1.2 \times 10^{-14} \text{ J}$$

$$\textcircled{11} \Delta V = \frac{\Delta U}{q}$$

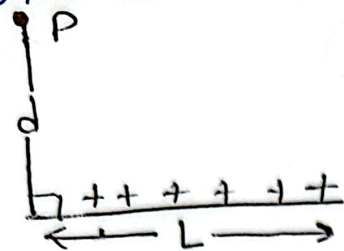
$$= \frac{-1.2 \times 10^{-14} \text{ J}}{-1.6 \times 10^{-19} \text{ C}}$$

$$= 75000 \text{ V}$$

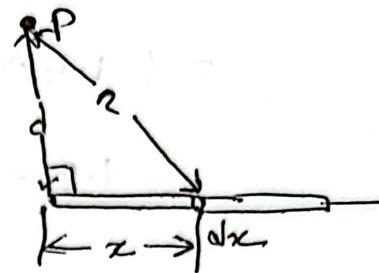
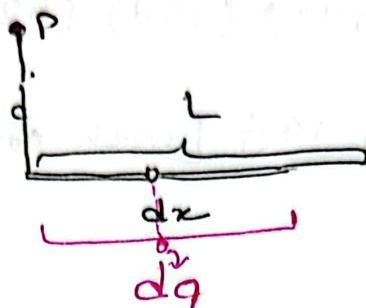
$$= 75 \text{ kV}$$

This tells us that the electric force does work to move the electron to a higher potential

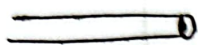
* Potential Due to Continuous Charge Distribution



→



Linear charge density λ



Q = total charge

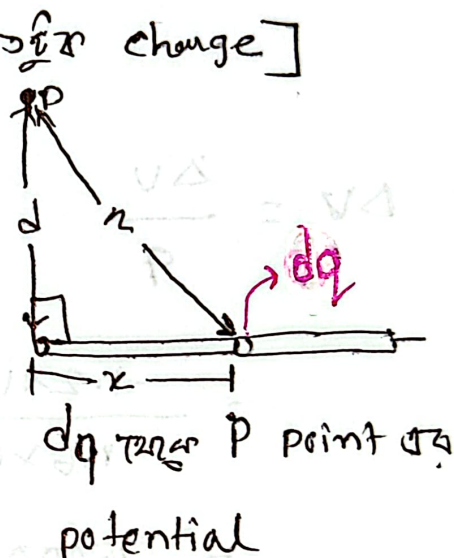
L = length

$$\therefore \lambda = \frac{Q}{L} \text{ [per unit length or } \lambda \text{ is charge]}$$

$$dq = \lambda dx$$

$$\therefore dv = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

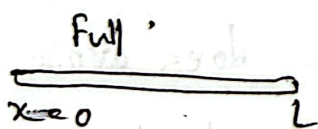
$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dx}{(x^2 + d^2)^{1/2}}$$



$$+ dv = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

$$* r^2 = x^2 + d^2$$

$$\therefore r = \sqrt{x^2 + d^2}$$



$$V = \int dv$$

$$= \int_0^L \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda}{(x^2 + d^2)^{1/2}} dx$$

$$= \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{dx}{(x^2 + d^2)^{1/2}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\ln(x + (x^2 + d^2)^{1/2}) \right]_0^L$$

$$\therefore = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(L + (L^2 + d^2)^{1/2}) - \ln d \right] \quad \boxed{\ln A - \ln B = \ln(A/B)}$$

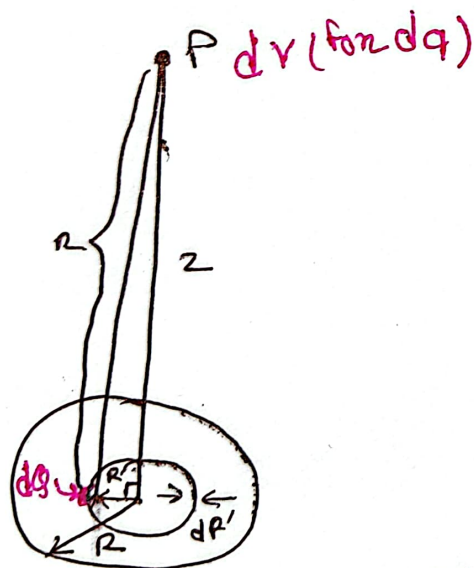
$$\therefore V = \frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{L + (L^2 + d^2)^{1/2}}{d} \right]$$

(derivation)

charge disk

$$dq = \sigma (2\pi R') (dR')$$

$$r = \sqrt{z^2 + R'^2}$$



$$dv = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma (2\pi R') (dR')}{\sqrt{z^2 + R'^2}}$$

$$\int dv$$

$$V = \int_0^R \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma (2\pi R') (dR')}{\sqrt{z^2 + R'^2}}$$

$$= \frac{\sigma}{2\epsilon_0} \int_0^R \frac{R' dR'}{\sqrt{z^2 + R'^2}}$$

$$= \frac{\sigma}{2\epsilon_0} \int \frac{R' dR'}{\sqrt{z^2 + R'^2}}$$

$$= \frac{\sigma}{2\epsilon_0} \int \frac{dx}{2\sqrt{x}}$$

$$= \frac{\sigma}{2\epsilon_0} \cdot \frac{1}{2} \int x^{-\frac{1}{2}} dx$$

$$= \frac{1}{2} \cdot 2 \cdot x^{\frac{1}{2}} \cdot \frac{1}{\sqrt{x}}$$

let $\rightarrow$ constant
 $z^2 + R'^2 = x$

$$2R' dR' = dx$$

$$R' dR' = \frac{dx}{2}$$

$$R' = 0$$

$$x = z^2$$

$$R' = R$$

$$x = z^2 + R^2$$

(46) (945) $\rightarrow$ pb

2014 15 7 = 0

$$\frac{PR}{a} = \frac{1}{\partial \partial P} = \text{vis}$$

(1000) 11 7



$$\frac{(46) (945) \rightarrow}{2014 15 7} \cdot \frac{1}{\partial \partial P}$$

$$\frac{(46) (945) \rightarrow}{2014 15 7} \cdot \frac{1}{\partial \partial P} \cdot \text{vis}$$

visibix

4614 47 20
 2014 15 7